

Exponential growth and decay



1

- a
- i Exponential growth
 - ii 60%
- b
- i Exponential decay
 - ii 80%
- c
- i Exponential growth
 - ii 105%
- d
- i Exponential decay
 - ii 22%

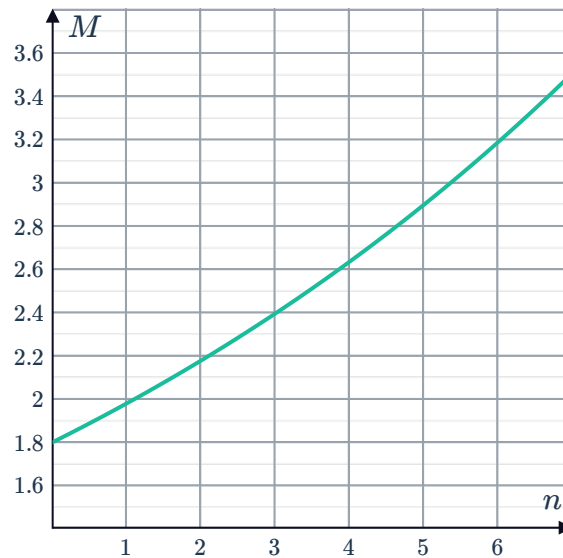
2

a $M = 2.4$

b

Months (n)	0	1	2	3	4	5	6
Mass (M)	1.8	2.0	2.2	2.4	2.6	2.9	3.2

c



3

- a 440 Hz b 262 Hz c $n = 73$

4

- a $y = 1300(1 + 0.08)^v$ b \$2806.60

5



6

a $y = 21 (0.5)^n$

b 0.66 yd

7

a $A = 300 (0.5)^{\frac{n}{20}}$

b 0.59 g

8

a 1105 km²

b 332 km²

9

a \$900.90

b \$819.82

c $A = 990 (91\%)^t$

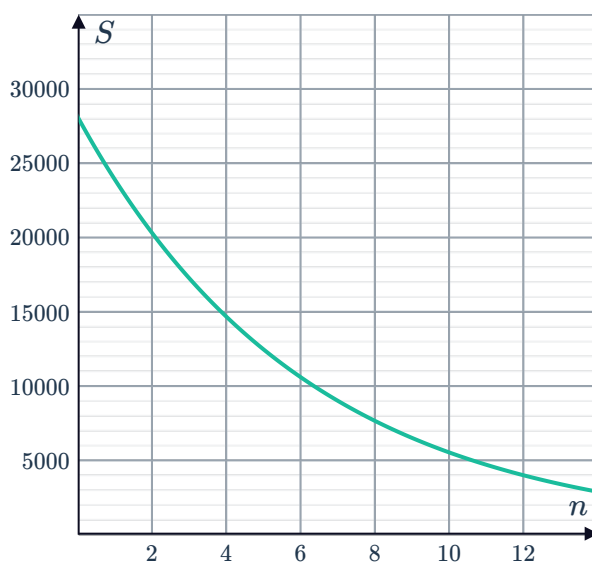
d \$319.25

10

a

n	2	4	6	8	10	12
S	20 230	14 616.18	10 560.19	7 629.73	5 512.48	3 982.77

b



c \$17 195.5

d 9 years

11

a 0.01

b $A = 38\,200 \times 0.99^n$

c \$35 964.54

d motorbike

12



a Decreasing

b

i No ii Yes iii Yes iv No

c The initial temperature of the substance

d $T = 145 \times 0.6^t$

e 0.88°C

f No

g Substance P

13

a $P = (0.85)^t$

b Maximilian

c C

d 13%

e No

Population growth

14

a 8 million

b

i 7130 million

ii 9286 million

15 155 432 people

16

a 1000 aphids

b 32 000 aphids

c 250 aphids

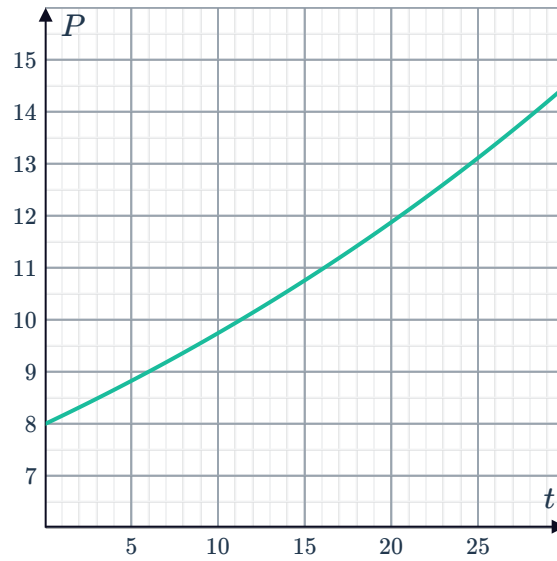
17



a

t	5	10	15	20	25	30
P	8.8	9.8	10.8	11.9	13.1	14.5

b



c 23 years

18

a 41 199 people

b 21 669 people

19 39 650 686 people

20 233 people

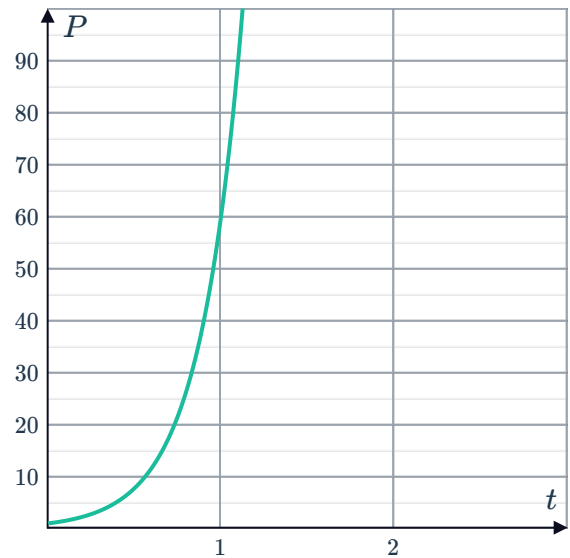
21 631 bacteria

22 6591 bees

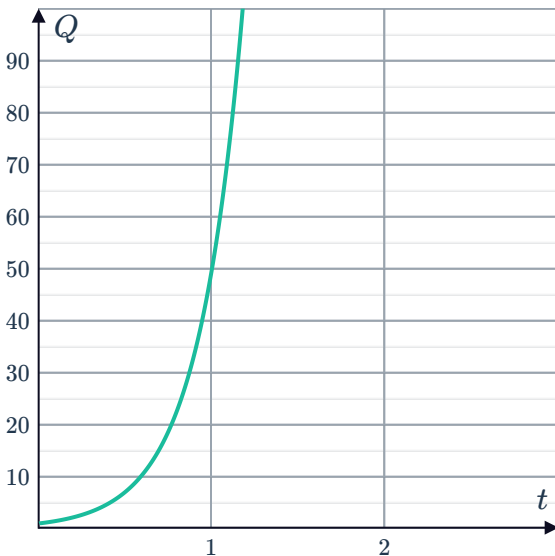
23 262 cows

24

a Faster



c



d No

25

a Yes, there will be a food shortage.

We can derive the equation $F = 3\,374\,000 (1.02)^n$ to represent the growth of the population, and the equation $P = 2\,410\,000 (1.022)^n$ to represent the number of people the food supply will feed.

We can see that the food production is increasing, but at a slower rate than the population's rate of increase, so the population will eventually exceed the amount that can be fed.

b We can draw the graph of both functions, using technology to find any points of intersection. If the y -value of the population is greater than the y -value of the population that can be fed for the same value of n then there will be a food shortage.



Compound interest

26 \$241.70

27

a \$35 730.90

b

n	1	2	3	4	5	6
A	32 190.00	35 730.90	39 661.30	44 024.04	48 866.69	54 242.02

c The value of the investment grows at an increasing rate.

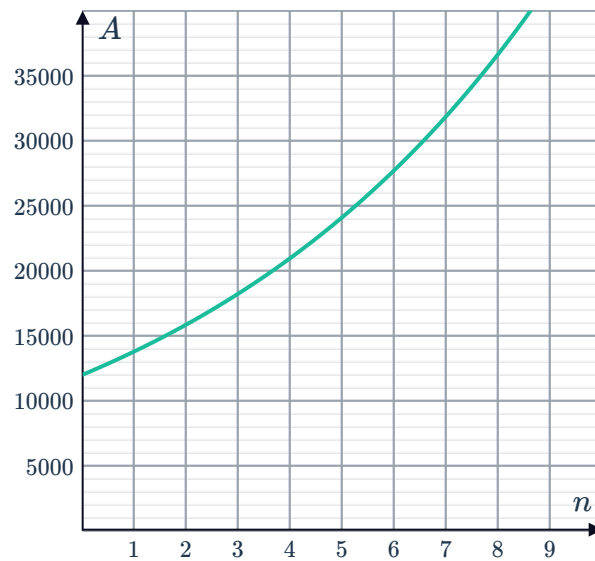
28

a \$15 870.00

b

n	1	2	3	4	5	6	7
A	13 800.00	15 870.00	18 250.50	20 988.08	24 136.29	27 756.73	31 920.24

c



d $n = 4$ years

29



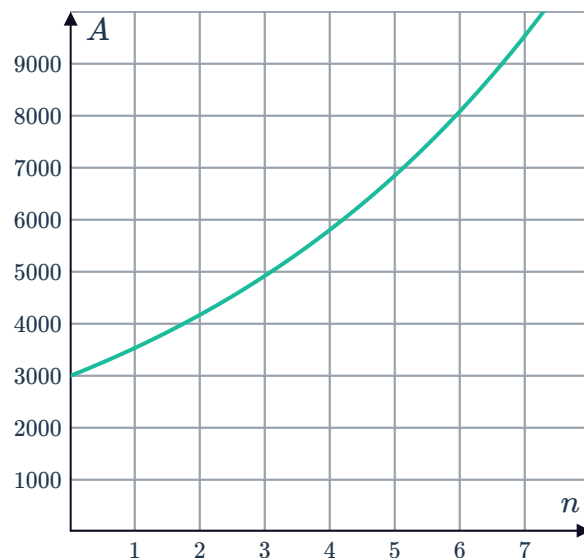
a $A = 3000 \times 1.18^n$

b \$8098.66

c

n	1	2	3	4	5	6	7
A	3540.00	4177.20	4929.10	5816.33	6863.27	8098.66	9556.42

d



30

a 0.02

b $A = 1000 \times 1.02^n$

c \$1126.16

31

a \$921

b 12

c \$35.43

32

a \$2500

b \$47 500

c \$20 000

d 10

33

a \$134 088.90

b \$24 088.90

34

a i $A = P(1.053)^t$

ii $A = P(1.0125)^{4t}$

b 5.09%

c Bank A



35

a \$1 614 774.00

b \$1 464 177.00

36

a Plan 1: \$1219.92
Plan 2: \$1291.17
Plan 3: \$1248.25

b Plan 2

37 \$0.29